

Facilities and Resources

TriadicFrameworks Research Initiative

Submitted to NASA High Priority Open-Source Science (HPOSS)

1. Overview

The TriadicFrameworks project requires only standard computational resources and open-source development tools. No specialized laboratory facilities, physical instrumentation, or high-performance computing clusters are necessary for the successful completion of the proposed work. All development, testing, and documentation activities can be performed using widely available software and cloud-based services.

2. Computing Resources

The project will utilize:

- modern development workstations capable of running standard open-source scientific and software tools,
- cloud-hosted repository services (GitHub) for version control, issue tracking, and continuous integration,
- cloud storage for documentation builds, example datasets, and reproducible workflows,
- open-source simulation environments for integration and testing.

These resources are sufficient for implementing dsrsp/0.1, RSM, vST, SLRP, and associated reference libraries.

3. Software and Tools

All required tools are open-source or freely available, including:

- programming languages (Python, JavaScript/TypeScript, or similar),
- documentation frameworks (Markdown, Sphinx, MkDocs),
- schema and validation tools (JSON Schema, YAML),
- Jupyter notebooks for reproducible workflows,
- open-source simulation engines for testing dsrsp/0.1 integration.

No proprietary software is required.

4. Data and Archival Infrastructure

The project will use:

- GitHub for public repositories and development workflows,
- Zenodo for DOI-assigned archival of specifications, datasets, and releases,
- GitHub Pages or equivalent for documentation hosting.

These platforms ensure long-term accessibility, transparency, and reproducibility.

5. Workspace and Environment

All work will be conducted in a standard remote development environment.

No physical laboratory space, specialized equipment, or institutional facilities are required. This makes the project cost-effective and fully aligned with HPOSS's emphasis on open, accessible scientific infrastructure.

6. Summary

The TriadicFrameworks project has access to all necessary computational, software, and archival resources to complete the proposed work. The project's requirements are modest, relying primarily on open-source tools and cloud-based services, ensuring efficient use of NASA funds and full alignment with open-science best practices.